Mahammad Salman

AI/ML Engineer | Full-Stack Developer

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SUMMARY

AI/ML and full-stack engineer building agent infrastructure: settlement rails, permission layers and reliability work for long-running systems. Upstream contributor to Nous Research Hermes Agent across the gateway, tool execution, state and desktop layers. MCA candidate in AI/ML Engineering and Data Science. Ships Python and TypeScript systems from model integration to production deployment.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Solidity, Move, Rust, Bash
AI and Machine Learning: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Hugging Face, LlamaIndex, RAG, LLM evaluation, feature engineering, data analysis
AI Agents: Model Context Protocol (MCP), Agent Client Protocol (ACP), x402, tool calling, memory stores, sandboxes
Web Development: React, Next.js, Node.js, FastAPI, Flask, Express, GraphQL, tRPC, REST APIs, WebSockets, Tailwind CSS
Data and Cloud: PostgreSQL, Redis, SQLite, MongoDB, pgvector, ClickHouse, Docker, GitHub Actions, Vercel, Cloudflare Workers, Linux, Terraform
Web3: Hardhat, Foundry, viem, wagmi, EVM, ERC-8004, ERC-7857
Testing: Pytest, Vitest, Playwright

EXPERIENCE

Open Source Contributor, Nous Research, Hermes Agent
Remote | Jul 2026 - Present

- 47 commit credits on the main branch of Hermes Agent (260k+ GitHub stars): 33 authored and 14 co-authored, plus 17 merged pull requests across Hermes Agent and its plugin repositories.
- Hardened gateway lifecycle in Python: durable startup-failure stamps, unprovable-ownership handling for restart loops, and liveness probe diagnostics.
- Bounded tool availability probes so a wedged check cannot hang an agent turn, and scoped git force-push approval patterns to the command segment.
- Improved SQLite state and cron reliability, including stable tool identity for display rows and failure stamps that keep healed runs visible.
- Published 10 standalone Hermes skills for monitoring and digests, covering TLS expiry, disk, public IP, releases and system vitals.

PROJECTS

PayMate | Next.js, TypeScript, Solidity, x402, PostgreSQL, Mistral

- Built non-custodial USDC invoicing and x402 pay-per-use billing for AI agents, live on GOAT mainnet with on-chain settlement verification.
- Shipped escrow contracts, ERC-8004 reputation credentials, LLM dispute arbitration and a replay-guarded, idempotent relayer ledger.

MediVault | TypeScript, Solidity, 0G, ERC-7857

- Built a private AI health vault with client-side encryption and ERC-7857 records, with a live mainnet contract.

GenLayer Adjudication Protocols | GenLayer, TypeScript, LLM consensus

- Shipped four protocols (Veritas, PolicyCourt, AeroSure, AgentPassport) that run LLM consensus inside contracts for fact-checking, disputes, insurance and bounties.

Cat Wallet | Next.js, TypeScript, Solidity, Hardhat

- Built a Telegram-native wallet with onboarding, swaps, staking and payments, including the contracts and liquidity pools behind it.

Client Systems | Next.js, TypeScript, PostgreSQL

- Delivered production software for two businesses: a travel company website, ERP and desktop billing app, and a hardware store site, admin console and billing POS.

EDUCATION

Master of Computer Applications (MCA), AI/ML Engineering and Data Science
Yenepoya University | In progress. Also studying robotics basics.
Bachelor of Computer Applications (BCA), Software Development and Web Design
Srinivas University | Completed